

1602000 – Electrical	
Spec Item	Title
1	General Information
1.1	Concept of Operations
2	Management and Administration
2.1	Definitions and Acronyms
2.2	Personnel
2.2.1	Certification and Training
2.3	Special Requirements
2.3.1	Safety
2.3.2	Arc Flash Program
2.4	References and Technical Documents
3	Recurring Work
3.1	Preventive Maintenance Program (PM)
3.1.1	High Voltage Systems Maintenance
3.2	Other Recurring Services
3.2.1	Photo Voltaic Systems Maintenance
4	Non-Recurring Work

1602000 - Electrical		
Spec Item	Title	Description
1	General Information	
1.1	Outline of Services	The intent of 1602000 Electrical is to specify the requirements for electrical power high voltage maintenance and photo-voltaic solar panel maintenance. Service Order response for Electrical are included in 1601000 Utility Management.

1602000 – Electrical		
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2	Management and Administration	
2.1	Definitions and Acronyms	Definitions and Acronyms are listed in J-1602000-01.
2.2	Personnel	The Contractor shall provide personnel with the qualifications, technical knowledge, experience and skills required to efficiently operate, maintain and repair the electrical power generation plant and transmission and distribution systems.
2.2.1	Certification and Training	The Contractor personnel responsible for operation, service, and repair of the electrical generation and distribution system shall be fully trained, qualified, and certified as electrical technicians in accordance with nationally recognized standards. The Contractor personnel operating and maintaining equipment shall understand and be knowledgeable of the SOPs that pertain to their area of assigned responsibility. Personnel under instruction operating equipment shall perform duties under the supervision of a qualified, knowledgeable individual. The Contractor's high voltage (HV) and low voltage (LV) personnel shall be qualified in accordance with NAVFAC P-604. The Contractor's high voltage (HV) personnel shall hold certificates of training CEI 11-27 for PES/PAV and CEI 78-17, and have designation letters CEI 11-27 for PES HV and CEI 78-17. The Contractor's low voltage (LV) personnel shall hold certificates of training CEI 11-27 for PES/PAV and have a designation letter CEI 11-27 for PES LV. The Contractor's HV supervisors or lead personnel shall hold certificates of training and designation letters for "Responsible of the Electrical LV and HV work " - " Preposto per lavori elettrici in bassa ed alta tensione". The Contractor's LV supervisors or lead personnel shall hold certificates of training and designation letters for "Responsible of the Electrical LV work " - " Preposto per lavori elettrici in bassa tensione". The Contractor shall submit proof of all certification, training, and licensing requirements in English per Section F. The Contractor shall provide documentation in English, that personnel have satisfactorily obtained training for the equipment and systems on which they will be working.
2.3	Special Requirements	
2.3.1	Safety	All the safety requirements of UFC 3-560-01 are applicable to this specification in addition to the safety standards contained in 0200000 Management and Administration.
2.3.2	Arc Flash Program	The Contractor shall implement and manage an Arc Flash program in accordance with UFC 3-560-1. The Contractor shall be responsible for identifying and properly addressing all Arc Flash hazards and maintaining all signage, labels, instructions, etc. for all medium and low voltage assets, regardless whether or not it is identified in J-sections, to include new assets not yet incorporated into the contract. The Contractor shall perform an Arc Flash Hazard Assessment of all medium and low voltage distribution assets within 90 days of contract start of the contract. The Arc Flash Hazard Assessment report shall be submitted to the Government for approval within 100 days after contract start. Within 30 days after Government approval of the report,

1602000 – Electrical		
Spec Item	Title	Description
		the Contractor shall install all labeling on the assets required by UFC 3-560-1. Labels shall be applied to the exterior of each panel and be of adequate materials to withstand environmental conditions.
2.4	References and Technical Documents	References and Technical Documents are listed in J-1602000-02.

1602000 – Electrical				
Spec Item	Title	Performance Objective	Related Information	Performance Standard
3	Recurring Work	The Contractor shall efficiently operate and maintain the electrical power transmission and distribution systems to safely produce, transmit and distribute reliable electrical power to ensure all electrical requirements are continually met.	The Contractor shall shut-down, restart, and perform operational checks on all equipment affected by both scheduled and unscheduled utility outages at no additional cost to the Government. The Electrical Power Transmission and Distribution Systems Description and Inventory is provided in J-1602000-03. The Contractor shall develop and submit a Critical Spare Parts Inventory List per Section F. The Contractor is fully responsible providing critical spare parts and for ensuring the critical spare parts list is updated and expanded as required and parts are available. The minimum critical Spare Parts that must be available is provided in J-1602000-04.	Electrical power is continuously and safely produced, transmitted and distributed and meets demand requirements. Critical spare parts are on-hand and available at all times.
3.1	Preventive Maintenance (PM) Program	The Contractor shall develop and implement a PM program for the electrical power transmission and distribution systems to ensure proper operation, to minimize breakdowns, and to maximize useful life.	The Contractor shall develop and submit a PM program, per Section F. The Contractor's PM program shall include UICAP requirements in 1601000 Utilities Management, NAVFAC P-603. The Contractor shall develop and submit a PM program plan, detailing the plans for each individual specification item, for Government review and approval per Section F. At a minimum, the PM program plan shall include: • PM management and technical approach • Procedures to evaluate equipment reliability and corrective actions • Job plans identifying each PM task • Qualification of the maintenance personnel. The Contractor is fully responsible for and shall perform any repairs, including replacement, discovered	Maintenance is accomplished in accordance with the Contractor's PM program and work schedule. PM is performed in accordance with manufacturers' recommended procedures and OEM standards.

1602000 – Electrical				
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			during scheduled maintenance work up to a total of \$3,000 per occurrence in direct material and labor cost under the recurring work portion of the contract. Incidental repairs work performed under maintenance are not considered a service order. Notification of repair work exceeding the incidental repairs limit shall be submitted to the KO within two hours of identification. A service order or non-recurring work task order may be issued for repairs exceeding the incidental repairs limits of liability. If non-recurring work is issued for repairs exceeding the incidental repair limits of liability, the Government will only pay for the portion of direct labor and/or direct material that exceeds the specified limits of liability. Example: If an individual occurrence of incidental repair requires \$3100 in direct labor and/or direct material cost, the Government may issue a service order or non-recurring work in accordance with the non-recurring work provisions of the contract for the \$100 in direct labor and/or direct material cost that exceeds the specified incidental repair limit of liability. The Contractor shall not use breakdown maintenance as part of the PM program. The PM program shall provide an economical approach using manufacturers' recommended procedures, OEM standards, and maintenance required to satisfy equipment warranties and keep installed equipment and systems in normal working condition. Excessive or repeated system or equipment breakdowns or deficiencies may indicate the need	

1602000 – Electrical				
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			to adjust or modify the Contractor's PM program. These changes will be made at no additional cost to the Government. The Contractor shall submit a monthly preventive maintenance schedule, unaccomplished maintenance report, repair status report, and summary of maintenance and repair accomplished under the various maintenance programs report, per Section F.	
3.1.1	High Voltage Systems Maintenance	The Contractor shall perform high voltage maintenance services on facilities, systems, and equipment to ensure they are fully functional and operational.	The electrical power distribution system includes all equipment and system components, such as, transformers, switchgear, high voltage transmission lines, and medium and low voltage distribution lines. Maintenance shall comply to CEI 78-17 Prior to performance of any high voltage maintenance services the Contractor shall co-ordinate with the PWD to switch electrical power off. PWD personnel are responsible for switching on and off electrical power. The Contractor shall perform all maintenance and repair work per applicable industry standards. At the minimum the standards shall include but not limited to all standards, requirements, and specifications; NETA MTS – 1997 "Maintenance Testing Specifications for Electrical Power Distribution Equipment and systems"; and NETA ATS – 1999 "Maintenance testing Specifications for Electrical Power Distribution Equipment and Systems". The Contractor shall provide all PPE necessary to perform services.	Systems and equipment are in an operable condition and function properly in accordance with specified standards. PM is accomplished per the Contractor's program and work schedule.

1602000 – Electrical				
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3.2	Other Recurring Services	The Contractor shall perform other recurring services on facilities, systems, and equipment to ensure they are fully functional and operational.	The Contractor shall develop and submit an Other Recurring Services Program, per Section F.	Systems and equipment are in an operable condition and function properly in accordance with specified standards. Other recurring services are completed in a timely manner.
3.2.1	Photo Voltaic Systems Maintenance	The Contractor shall perform maintenance on photo-voltaic panel systems and equipment to ensure proper operation, to minimize breakdowns, and to maximize useful life.	The inventory of the Photo Voltaic Panel Systems and Equipment is provided in J-1602000-03. The Contractor shall clean PV panels and verify integrity twice per year For PV strings, the Contractor shall verify tension and current, verify insulation, verify integrity, validate and check tightening of electrical wires and clamps, twice per year. For Inverters (manager and modules), the Contractor shall clean, verify integrity, validate and check tightening of the electrical cables, verify insulation of all connection, check using a thermal vision cameras the temperatures of connection during the exercise twice per year. For Electric panels,; the Contractor shall cleaning, verify integrity of all circuit breakers, verify integrity, validate and check tightening of clamps and electrical cables, verify short circuit current for each breaker, verify GFI (ground fault interrupter), frequency: twice per year. For Support structure for PV panels, the Contractor shall check of bolts tightening, verify structure integrity and presence of rust, verify grounding continuity, Following each PM activity, the Contractor shall provide a	Photo Voltaic Systems and equipment are in an operable condition and function properly in accordance with specified standards. Maintenance is accomplished per the Contractor's program and work schedule.

1602000 – Electrical				
Spec Item	Title	Performance Objective	Related Information	Performance Standard
			Photovoltaic Systems Maintenance Report per Section F.	

1602000 – Electrical				
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4	Non-Recurring Work	Non-Recurring work may be ordered utilizing FedMall in accordance with Section H or on a task order in accordance with the PROCEDURES FOR ISSUING ORDERS clause in Section G. The order will specify the exact locations and types of work to be accomplished. The period of performance will be specified in each order.	Refer to Non-Recurring work ELINs for task listings, descriptions and related requirements. All periods of performance are measured from issue date of order to acceptance of the work.	Performance Standards for Non-Recurring work will be the same as those in Spec Item 3 where applicable.